

Electromagnetic Spectrum

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	The electromagnetic spectrum runs from radio to gamma rays; as wavelength decreases, frequency and energy increase. Visible light is only a small portion.
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Round 1 - Retrieve It

1. List the spectrum from longest wavelength to shortest.

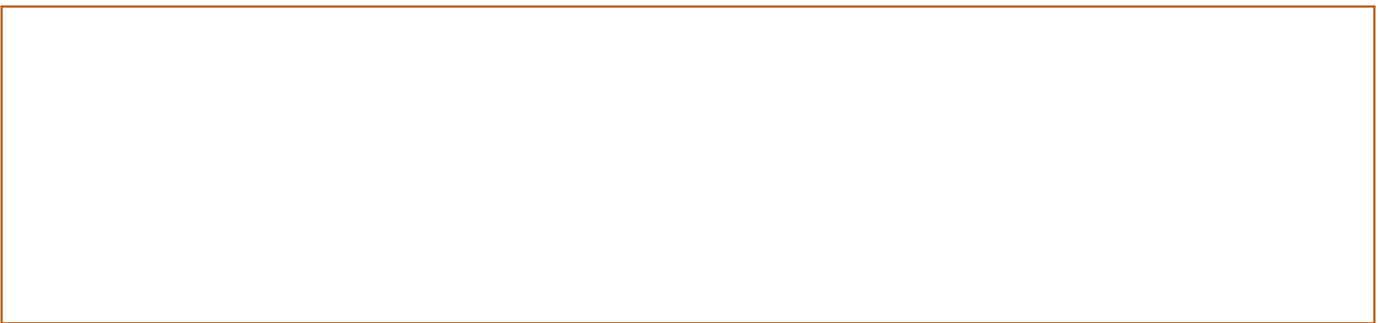
2. As frequency increases, what happens to wavelength?

3. As frequency increases, what generally happens to photon energy?

4. Is visible light most of the electromagnetic spectrum?

Round 2 - Sketch It

Draw a horizontal spectrum bar from radio to gamma. Add arrows showing wavelength decreases while frequency and energy increase.



Electromagnetic Spectrum - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which has longer wavelength: radio waves or X-rays?

2. Which has higher frequency: ultraviolet or infrared?

3. Match the use: thermal imaging/remote controls, bone imaging, communication broadcasts.

Round 4 - Reason from Evidence

Compare radio waves and gamma rays using wavelength, frequency, and energy.

Electromagnetic Spectrum - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Electromagnetic Spectrum in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect visible light to the full spectrum. Why can two electromagnetic waves behave differently even though both are electromagnetic radiation?

E**Extend**

A new detector measures radiation with a shorter wavelength than visible light but longer than X-rays. Which region is a likely possibility?

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Electromagnetic Spectrum

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.
2. It decreases.
3. It increases.
4. No. It is a small portion.

Sketch It - look for

Correct order with wavelength arrow opposite frequency/energy arrows.

Apply It

1. Radio waves.
2. Ultraviolet.
3. Infrared; X-rays; radio waves.

Reason from Evidence - look for

Radio: longest wavelength, lowest frequency/energy end. Gamma: shortest wavelength, highest frequency/energy end.

ACE - Extend look-for

Ultraviolet is the most direct answer in the standard order; depending on exact wavelength, boundary regions may vary.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.